NIST prime-curve trapdoor slots (P-192 … P-521), status after round 0

S2 prime p
Solinas form, open choice

→

S7 seed
84 public preimages: 0 hits

→

S3 seed → b
root sign = twist on 4 of 5

→

S1 model (p, a, b)
the only sub-exp slot

→

S4 base point G
never first valid x

Isogeny-invariant (public, checkable)

S6 / S5 / S2 census, executed

n prime, cofactor 1 (5/5)
embedding degree k > 1000 (5/5)
$|D| = 4p - t^2$: no square part ≤ 10^6
except 3^3 on P-224; no square residue
twist orders factored to 2^22 rho:
P-384 prime; P-224, P-192 residues open
n ± 1: large residues (5/5)
any weakness here cannot be hidden by an isogeny

Model-dependent (can be isogeny-hidden) — pending rounds

S1a cover route (E1a, toy p 2^20–2^28)
predict slope 2 − 2/g > 0.55; falsify if below

S1b class-group map (E1b, 2^16 curves)
predict 0 curves with n | h(O); Lagrange bound

S1c factor-base carrier (E1c, 136 vertices)
predict Semaev PDP cost constant on isogeny class

S1d hiding asymmetry (E1d, L ≤ 32)
predict known-target predicate-only recovery, ratio > 2^10
Teske 2006 template; no F_p instance known

S3 grind density (E3, 40-bit p, 2^24 seeds)
predict public classes ≤ 2^-16; root-sign unbiased

S4a rigid-G test (E4a)
predict no deterministic rule reproduces G

S4b protocol audit (E4b, literature)
Dual_EC pattern table per deployed protocol

S5 twist completion (E5, ECM)
predict P-224 largest twist prime ≈ 2^117
S2 matched-rho control (E2): equal iteration counts

Not publicly testable

S8 a better generic algorithm (a capability, not a parameter trapdoor; docs/ic/BOUNDARY_TARGETS.md regime C) · S7 seed = SHA-1(secret ‖ counter): only a preimage closes it

executed, no anomaly · executed, unexplained freedom recorded (not evidence of grinding) · registered, pending · not testable

Round 0 evidence: examples/nist_slot_census.rs → census.md / census.json (sha256 in PROTOCOL.md §3). Curves: CurveParams::p192 … p521 (FIPS 186-4 D.1.2); seeds SEC 2 §2.4.
Principle: a weakness that is a function of #E, t, k or the square-free part of $t^2 - 4p$ is constant on the isogeny class and cannot be hidden; only a model-dependent property (a cover, a factor-base structure, a GHS-type descent) can be hidden by an isogeny walk, and over F_p no such property with a sub-exponential solver is known.

Proposed and pending items are drawn in blue and make no measured claim. No weakness in any NIST curve is claimed or found.